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Task Description:	Simply put, individuals who do not get enough sleep do not perform well when awake. Although physical rest is an important part of the benefit gained from sleep, adequate sleep is essential for optimal brain function. Not surprisingly, inadequate sleep is a great concern for astronauts during spaceflight missions. Astronauts often get less nightly sleep than they should (< 7 hours) and may be forced to get little sleep for prolonged periods. Unfortunately, taking sleeping pills is often counterproductive. In addition to other unwanted side effects, like addiction and increased tolerance with prolonged use, sleep medications often lead to daytime grogginess and a lack of mental alertness, the opposite of the mentally refreshed feeling sleep is supposed to provide. Our group has spent over a decade developing a technique to optimize sleep without using medications. We designed a system that capitalizes on two of the most important things the brain needs to do during sleep - produce slow waves and monitor the environment. Slow waves are the dominant rhythm of the brain when it is in its deepest and most restorative sleep state. When the brain goes offline during sleep, slow waves work to actively restore the brain in preparation for the next day. However, the act of going offline during sleep is dangerous for the brain because it can no longer do one of its primary functions - protect the body. Therefore, even during deep sleep, the brain subtly monitors the surrounding environment, most effectively by listening for sounds. The device we developed takes advantage of this fact and actively monitors the brain during sleep to reliably detect slow waves. When slow waves are detected, auditory stimuli are played at a frequency that results in a robust increase in slow waves, the most important restorative rhythm of sleep. However, the overall benefits of enhancing sleep slow waves with auditory stimulation has not been adequately tested. Largely, this is because an adequate system for nightly stimulation has not been available. While several research groups have recently demonstrated improved memory retention in the laboratory after playing appropriately timed sounds during slow wave sleep, these effects were demonstrated using a different stimulation protocol than the one we developed and typically did not show an overall increase in sleep slow waves across the night. This suggested that other than improved memory, the stimulation might not affect the other known restorative benefits of sleep. Assessing the cognitive effects of tone-enhanced sleep is also challenging in the laboratory due to the variability in sleep, between different people, and even night-to-night within the same person. On any given night in the laboratory, sleep might be better and subsequent cognitive performance might improve but this will have nothing to do with the stimulation and will have more to do with chance. Maximizing the sleep enhancing benefit of a system means adapting to and accounting for an individual and his/her changing sleep patterns and testing the subject for longer periods of time, not just a few nights. The ability to enhance sleep with a wearable device that is comfortable, easy to use, and continuously adapts to an individual is a truly exciting prospect. The proposed study will be the first to assess the ability of different ways of playing auditory tones during sleep to enhance slow waves and improve cognitive performance over a long-term period (8 weeks). We will record and stimulate subjects in their home environment using the recently developed, commercially available wearable device and test subject performance throughout each day using a short-term test battery developed particularly for astronauts. This proposal represents the vital first steps towards optimizing auditory stimulation during sleep to enhance long-term cognitive performance using a device that is easily translatable to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	SmartSleep is a recently developed wearable electroencephalogram (EEG) device that is comfortable and easily translatable for spaceflight, with high sleep staging accuracy and capable of customizable closed-loop auditory stimulation during sleep. This study will be the first of its kind to directly evaluate the ability of various auditory stimulation protocols to enhance sleep and cognitive function in an ecological, at-home sleep environment. Prior to now, nothing existed which would make this type of study feasible. If successful, this study would demonstrate ways that auditory stimulation could be optimized in space to improve astronaut performance and propose how stimulation should be tailored to the individual and the particular mission tasks.
Task Progress:	[Ed. note May 2020: Report submitted by TRISH to Task Book in March 2020; covers reporting as of December 2019.] During spaceflight missions, astronauts may not have adequate opportunity for sleep, yet they are required to maintain alertness for extended periods. As demonstrated by several well-controlled studies, even modest sleep restriction takes a heavy toll on reaction time, motivation, attention, memory, endurance, and judgment. Therefore, when opportunities for sleep are available, it is essential that sleep is as efficient as possible. Accumulating evidence suggests many of the benefits of sleep are associated with sleep slow waves. Slow waves reflect the near-synchronous alternation by millions of neurons between depolarized upstates and hyperpolarized downstates. These alternations occur at a rate of about 1 Hz in the brain and are thought to mediate the restorative effects of sleep on brain networks and cells. Sleep slow waves support system and synaptic consolidation by promoting specific patterns of neuromodulatory and electric activities. Subjective sleep quality is also related to the size and number of slow waves. Unfortunately, pharmacologic approaches to improve sleep efficiency by increasing slow waves have been relatively unsuccessful. Moreover, these pharmacological agents are often associated with residual side effects during wakefulness and are likely to remain in their system if an astronaut is unexpectedly awoken from sleep needing to function. Recent data from several different laboratories have now confirmed that low intensity (sub-arousal) auditory tone stimulation when given regularly at 1 Hz during non-REM (non-rapid eye movement) sleep (the approximate rhythm that the brain is naturally oscillating), can consistently enhance slow waves and cognitive performance non-pharmacologically. This type of stimulation is not only effective, but also it is safe, controllable, and non-obtrusively administered during sleep. Several groups have shown positive next-day effects on memory retention after auditory enhanced sleep. Despite these common successes though, the precise types of auditory stimulation protocols used by different laboratories varies substantially. Assessment of the long-term impact on sleep and performance of auditory sleep enhancement paradigms requires an ecological, user-friendly, and flexible platform for stimulation comfortable enough to be worn routinely. In this study, we are using a sleep wearable prototype (SmartSleep), developed by our group in collaboration with Philips Sleep and Respiratory Care, that monitors sleep EEG in real-time. The device enhances slow waves by delivering auditory stimulation in the form of 50-millisecond tones during non-REM sleep in a closed-loop fashion dynamically modulated by sleep depth. Twenty-four subjects (healthy, aged 30-55) will use SmartSleep at home for 8 weeks. Three different auditory stimulation protocols will be tested for two-week periods with the order randomly assigned and blinded to the subjects: (1) FIXED ITI, 1 second continuous inter-tone interval; (2) BLOCK 5 seconds ON versus 5 seconds OFF, 1 second inter-tone interval; (3) IN-PHASE continuous stimulation with tones timed to be delivered during each upstate of the slow wave. Subjects will perform the Psychomotor Vigilance Test (PVT) along with a series of validated behavioral questions that assess sleepiness, physical fatigue, and mental tiredness 30 minutes after awakening each day and will fill out the Karolinska Sleepiness Scale three times throughout the day. In addition, subjects will perform a full Cognition test battery one hour before sleep each night. A no-stimulation (SHAM) week will occur between each of the 2-week stimulation periods. During this week, participants will continue to perform the daily

	neurocognitive testing. The aims of the study are to (1) confirm that sleep and cognitive function can be enhanced using closed-loop auditory stimulation during sleep in a group of astronaut-like subjects in an ecological setting using objective and subjective sleep measures and a validated cognitive test battery; (2) determine the extent to which various closed-loop auditory stimulation protocols (FIXED ITI, BLOCK, IN-PHASE adjustable) differentially affect various aspects of sleep and cognition in individuals; (3) explore whether an individual's previous night sleep EEG metrics can predict cognitive performance in a group of astronaut-like subjects; and (4) assess any potential adverse effects of long term slow wave sleep enhancement.
Bibliography Type:	Description: (Last Updated: 08/04/2022)
Abstracts for Journals and Proceedings	Riedner BA, Garcia-Molina G, Flaherty BL, Jones SG, Basner M, Dinges DF, Tononi G. "OASIS: Optimizing Auditory Stimulation to Improve Cognitive Performance Performance Using SMARTSLEEP." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020